# 11 · Ad exposure is self-selected — the true effect (instrumental variables · CausalPy)

July 13, 2026

> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

**The business decision.** People who see our ads convert more — but the customers who *end up* seeing our ads are also the ones already leaning toward us (they browse, they search, they get retargeted). So **exposure is self-selected**, and a plain regression of sales on exposure is biased **upward**: it credits the ad for enthusiasm that was already there. We want the *true* causal effect of an extra exposure so we can set the right **budget** — the most we can pay per exposure and still come out ahead.

### The problem in one word: endogeneity

When the treatment (ad exposure) is **correlated with the hidden drivers of the outcome** (a customer's unobserved intent), we say exposure is **endogenous**, and no amount of controlling for what we *measured* fixes it — the confounder (intent) is unmeasured. Notebook 05's adjustment tools are powerless here.

### The fix: an instrument

An **instrumental variable (IV)** is a third variable $Z$ that nudges the treatment *without* affecting the outcome any other way. Here $Z$ is a randomized **encouragement** — e.g. a serving-priority lottery that makes some customers *more likely* to be shown the ad. Because $Z$ is randomized, the slice of exposure it drives is as-good-as-random, uncontaminated by intent. IV isolates that clean slice and reads the effect off it. A valid instrument needs **four** things:

- **Relevance** — $Z$ actually moves exposure (this one we *can* test: the **first-stage F-statistic**; $F < 10$ is a "weak instrument" danger sign).
- **Exclusion** — $Z$ affects sales *only through* exposure (the lottery itself sells nothing). Untestable from data — defended on design, and stress-tested in Step 6.
- **Exogeneity** — $Z$ is independent of the hidden confounder (guaranteed here by randomization).
- **Monotonicity** — the encouragement never pushes anyone *away* from exposure (no "defiers": nobody is so annoyed by the nudge that they block ads in response). Untestable, but usually plausible for a gentle encouragement — and it is what buys the LATE interpretation below (derived in Step 3).

### What IV recovers: a LATE

IV does **not** give the average effect over everyone. It gives the **LATE** (*Local Average Treatment Effect*) — the effect *for the "compliers,"* the customers whose exposure the encouragement actually changed. Always-exposed loyalists and never-exposed sceptics are silent in an IV estimate; report the scope accordingly.

> **On real data.** IV needs a genuine instrument, which is the hard part. Good marketing instruments: randomized **encouragement / intent-to-treat** designs (ship the nudge to a random half), ad-auction or delivery randomness, or a **cost shifter** for the price-elasticity problem in notebook 03. If you can randomize the *encouragement* (not the exposure itself), you have a clean instrument. Notebook **11b** (`11b_endogenous_exposure_real_data.ipynb`) runs this exact playbook on the real Criteo uplift data, where the randomized assignment instruments actual ad exposure — the real-world leg of this encouragement design.

## 2 · Simulate a ground truth

The **true** exposure→sales effect is **€15**. An unobserved `engagement` drives both exposure and sales (the confounding), so naive OLS overstates it. A randomized `encouragement` (the instrument) nudges exposure without touching sales directly.

**The data-generating model** — exactly what `dgp.iv_ad_exposure` implements (defaults & seed in `src/cmp/dgp.py`). $n = 3000$; **unobserved** intent $U$ = engagement $\sim \mathcal{N}(0,1)$; randomized encouragement $Z \sim \text{Bernoulli}(\frac{1}{2})$; $\sigma(z) = 1/(1+e^{-z})$:

$$\begin{aligned} X = \text{exposure} &\sim \text{Bernoulli}(\sigma(0.3 + 1.1\,Z + 0.8\,U)) && \text{(first stage)} \\ Y = \text{sales} &= 50 + 15\,X + 12\,U + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,\,6^2) && \text{(true effect: the 15).} \end{aligned}$$

$U$ appears in **both** equations — $0.8\,U$ pushes exposure and $12\,U$ pushes sales — that is the self-selection, and since $U$ is unobserved no regression adjustment can fix it (OLS is biased upward). All **four** IV conditions are visible by construction: **relevance** ($1.1 \neq 0$ — testable, the first-stage F below), **exclusion** ($Z$ appears nowhere in the sales equation — untestable in real data, stress-tested in step 6), **exogeneity** ($Z$ is a fair coin flip, drawn independently of $U$ — randomization), and **monotonicity** ($1.1\,Z$ only ever *raises* the exposure probability — no defiers).

```
TRUE exposure->sales effect €15  *  first-stage: exposure 56%->77% (shift +21
pp), F = 156
```

| | encouragement | ad_exposure | sales |
|---|---|---|---|
| 0 | 1.0 | 0.0 | 63.748992 |
| 1 | 1.0 | 1.0 | 64.636639 |
| 2 | 1.0 | 1.0 | 46.507627 |
| 3 | 0.0 | 1.0 | 75.599770 |
| 4 | 0.0 | 1.0 | 70.577754 |

## 3 · Identify — a valid instrument, and what IV recovers

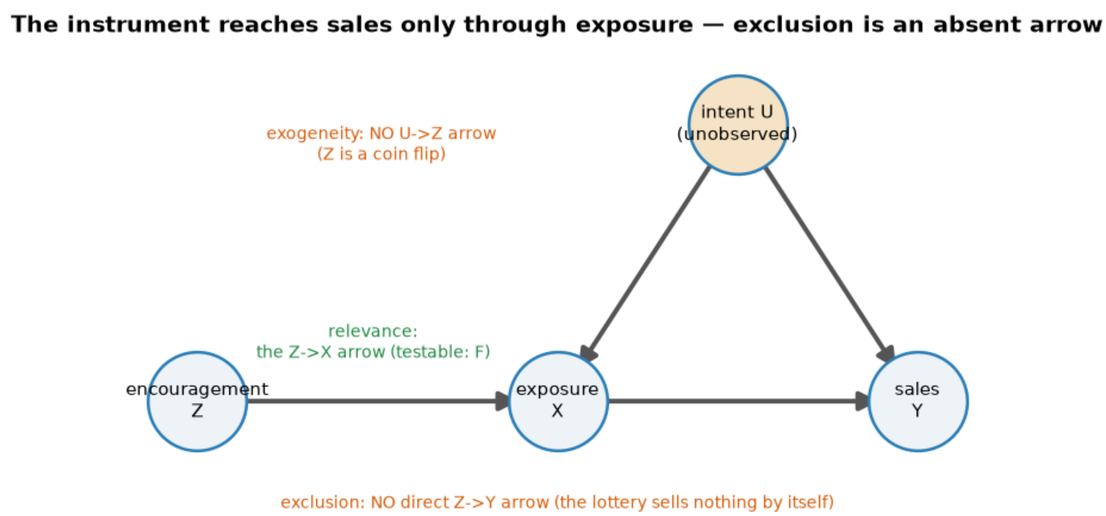
Exposure is **endogenous** — it is correlated with the outcome's hidden driver, Cov(exposure, $12U + \varepsilon) \neq 0$ (the $U$ term specifically, since $\varepsilon$ is independent structural noise) — so OLS is biased. The four requirements from the intro, now each named, formalized, and argued in marketing terms:

- **Relevance** — $\pi \neq 0$ in the first stage $X = b_0 + \pi Z + u$: the encouragement genuinely moves exposure. *Testable* — the first-stage F printed in Step 2 (F = 156 here; §5c defines F, explains the F < 10 rule, and shows what happens as an instrument decays).
- **Exclusion** — the encouragement has **no arrow of its own into sales**. In potential-outcomes notation, writing $Y_i(z, x)$ for customer $i$'s sales if we *set* encouragement to $z$ and exposure to $x$:

$$Y_i(z, x) \;=\; Y_i(x) \qquad \text{for all } z \in \{0, 1\} :$$

holding exposure fixed, flipping the lottery changes nothing. *Untestable from data* — defended on design (a serving-priority lottery sells nothing by itself; an encouragement *email that shows the product* would violate it), and stress-tested in Step 6. - **Exogeneity / independence** — $Z \perp (U, \varepsilon)$: the instrument is unrelated to the hidden drivers of exposure and sales. Holds here **by randomization** of the lottery. - **Monotonicity** — $X_i(1) \geq X_i(0)$ for every customer $i$, where $X_i(z)$ is $i$'s exposure when the encouragement is set to $z$: the nudge never pushes anyone *away* from the ad (no defiers). *Untestable*; plausible for a gentle encouragement, and it is precisely what turns the IV ratio into an interpretable LATE — the derivation below uses it to kill one term.

The DAG states the structural part in one picture — and teaches that two of the assumptions are **missing arrows**:




**Read the DAG by its absences.** The unobserved intent $U$ points into both exposure and sales — that fork is the endogeneity OLS trips over, and no arrow into $U$ can be measured away. The instrument $Z$ has exactly **one** outgoing arrow, into exposure: exclusion and exogeneity are not things $Z$ *does*, they are arrows $Z$ *does not have* — which is why they cannot be tested from data, only designed for. Relevance is the one arrow $Z$ must have, and the only assumption the data can check.

**Who is in the estimate? The four customer types.** With $X_i(z) \in \{0, 1\}$ the exposure customer $i$ would end up with under encouragement $z$, two potential exposures give four types — the IV analogue of notebook 01's Persuadable / Sleeping-Dog grid:

| | $X_i(1) = 1$ — exposed if encouraged | $X_i(1) = 0$ — unexposed if encouraged |
|---|---|---|
| $X_i(0) = 1$ — exposed anyway | **Always-taker** — brand loyalist who seeks the ad out regardless; the lottery changes nothing | **Defier** — so annoyed by the nudge they block the ad they'd otherwise have seen. *Ruled out by monotonicity* |
| $X_i(0) = 0$ — unexposed otherwise | **Complier** — the lottery tips them into seeing the ad. *The only type IV speaks for* | **Never-taker** — ad-blocker user, barely online; unreachable either way |

**The LATE theorem (Imbens–Angrist), in two lines.** Decompose the **ITT** (intent-to-treat) effect on sales — the average sales difference between encouraged and not-encouraged customers, which *is* the reduced form — over the four types. Randomization of $Z$ makes encouraged vs not comparable; **exclusion** says sales can only react to $Z$ if exposure changed, which silences always- and never-takers ($X_i(1) = X_i(0)$); **monotonicity** deletes the defier term. What survives is the compliers:

$$\underbrace{E[Y \mid Z{=}1] - E[Y \mid Z{=}0]}_{\text{ITT on sales (reduced form)}} \;=\; E\big[\big(Y_i(1){-}Y_i(0)\big)\big(X_i(1){-}X_i(0)\big)\big] \;=\; \Pr(\text{complier})\cdot\underbrace{E\big[Y_i(1) - Y_i(0) \mid \text{complier}\big]}_{\text{LATE}}.$$

The same decomposition applied to exposure itself gives $E[X \mid Z{=}1] - E[X \mid Z{=}0] = \Pr(\text{complier})$ — **the first stage *is* the complier share** (the +21 pp printed in Step 2: about a fifth of these customers are compliers). Divide the two:

$$\text{LATE} \;=\; \frac{E[Y \mid Z{=}1] - E[Y \mid Z{=}0]}{E[X \mid Z{=}1] - E[X \mid Z{=}0]} \;=\; \frac{\text{reduced form}}{\text{first stage}} \;=\; \hat{\beta}_{\text{Wald}}.$$

So IV does **not** estimate the average effect over everyone: always-takers and never-takers cancel out of the numerator and are literally absent from the estimate — the LATE is the effect *for the customers the encouragement can move.* CausalPy fits the Bayesian version of this ratio: a joint model of (exposure, sales) with correlated errors, *estimating* the endogeneity $\rho$ rather than assuming it away.

> **Modeling note.** Because `ad_exposure` is binary (0/1), this joint MvNormal models the exposure first stage as a **linear-probability (Gaussian)** equation, so the two-equation model is an approximation — the exposure equation's posterior *width* shouldn't be read at face value; the reliable read-outs are its point estimate and the endogeneity $\rho$. §4b makes this misfit visible on purpose.

## 4 · Estimate — OLS, reduced form, and Bayesian IV

Three regressions tell the whole story, and seeing them side by side demystifies what IV actually does:

- **OLS** — regress sales on exposure directly. This is the **biased** number, inflated by self-selection.
- **Reduced form** — regress sales on the *instrument* (encouragement → sales). This captures only the effect that flows through the clean, randomized channel.
- **First stage** — regress exposure on the instrument (encouragement → exposure): how much the nudge moves exposure.
- **Wald / IV ratio** — reduced form ÷ first stage. The intuition: the instrument raised sales by *this* much and raised exposure by *that* much, so each unit of (instrument-driven) exposure is worth their ratio. That ratio *is* the causal effect. The **Bayesian IV** below is the same idea fit jointly with uncertainty (and it estimates the endogeneity $\rho$ rather than assuming it away).

**The fitted models, in symbols.** The three frequentist point estimates:

$$\text{OLS: } Y = a_0 + a_1 X; \qquad \text{first stage: } X = b_0 + \pi Z; \qquad \text{reduced form: } Y = c_0 + \delta Z; \qquad \widehat{\beta}_{\text{Wald}} = \frac{\delta}{\pi};$$

and the Bayesian IV fits the first stage and the outcome equation **jointly**, with correlated errors:

$$\begin{pmatrix} X_i \\ Y_i \end{pmatrix} \sim \mathcal{N}_2\left( \begin{pmatrix} b_0 + \pi Z_i \\ \beta_0 + \beta X_i \end{pmatrix}, \, \Sigma \right), \qquad \Sigma = \begin{pmatrix} \sigma_X^2 & \rho\,\sigma_X \sigma_Y \\ \rho\,\sigma_X \sigma_Y & \sigma_Y^2 \end{pmatrix},$$

where the off-diagonal $\rho$ **is** the endogeneity: the correlation between the two equations' errors that the hidden $U$ induces and that OLS ignores. Priors: weakly-informative $\mathcal{N}(0, 50^2)$ on the regression coefficients (the code comment explains why the library default is too tight here) and an **LKJ prior** on $\Sigma$ — a standard weakly-informative prior over correlation matrices that lets the data decide how strongly the two equations' errors move together (i.e. how big $\rho$ is), rather than fixing it in advance.

**Predict the OLS bias before running the regression.** Because we built the DGM, we can compute exactly how wrong OLS will be — before it is wrong. With $Y = 50 + 15X + 12U + \varepsilon$, the OLS slope of $Y$ on $X$ converges to the planted effect **plus** whatever part of the error leaks in through the treatment (the **omitted-variable-bias** formula, specialized to this DGM):

$$\hat{a}_1 \ \xrightarrow{p}\ 15 \ + \ \frac{\mathrm{Cov}(X,\ 12U + \varepsilon)}{\mathrm{Var}(X)} \ = \ 15 \ + \ 12\,\frac{\mathrm{Cov}(X, U)}{\mathrm{Var}(X)},$$

since $\varepsilon$ is independent noise. Even the sign is knowable in advance: the first stage loads on $U$ with $+0.8$, so $\mathrm{Cov}(X, U) > 0$ and OLS must be biased **upward**. The code below re-draws the same simulation *retaining the normally-unobservable* $U$, evaluates this formula, and only then runs the regressions — theory should call the OLS number before the regression prints it. The same $\mathrm{Cov}(X, U)$ has a second life: it is what the Bayesian joint model estimates as a **positive error correlation** $\rho$ between the exposure and sales equations.

```
OVB theory (computed before the regression):
  OLS ->p 15 + 12*Cov(X,U)/Var(X) = 15 + 8.5 = €23.5
OLS (biased)      €23.7  - the bias is computable self-selection, not bad luck
Reduced form (Z->Y) €3.5  /  first stage (Z->D) 0.21  =  Wald/IV ratio €16.5
Bayesian IV        €17.7  (true €15)  90% credible interval [€14.3, €21.1]
IV convergence: max r-hat 1.010 - min ESS 903 - divergences 0
Estimated endogeneity  rho(exposure err, sales err) = +0.22 [90% +0.10, +0.33]
  - nonzero rho is why OLS (€23.7) sits above IV (€17.7); it is the Bayesian
  image of the same Cov(X,U) the OVB formula priced at +€8.5.
```

**Reading the sampler's two health checks.** The Bayesian IV is fit by simulation — **MCMC**, the algorithm that draws samples from the posterior — so PyMC prints two numbers that say whether the sampler actually converged, and it will warn loudly if they look loose. **R-hat** compares the variance *within* each MCMC chain to the variance *across* chains; it should sit at ≈1.00, and **≤ 1.01 is the usual pass bar**. **ESS** (effective sample size) is roughly how many *independent* draws the autocorrelated chain is worth — a few hundred is ample for a posterior mean or interval. Under this notebook's FAST teaching profile the chains are deliberately short (only 200 draws × 2 chains to keep it quick), so R-hat can drift to ~1.04 and ESS can fall toward ~80, tripping PyMC's *"problems during sampling"* and *"effective sample size ... smaller than 100"* notices. Those are **artifacts of the small draw count, not a broken estimate** — they clear in a FULL run, and they flag *how well we sampled the posterior*, never whether the IV logic is right. (The `90% credible interval` printed on the same line is the Bayesian analogue of a confidence interval: given the model and priors, there is a 90% posterior probability the true effect lies inside it — which is what lets us later say "the whole posterior sits above €10." The real backstop that the number is trustworthy is the multi-seed recovery-and-coverage check in §5b, not any single run's sampler line.)

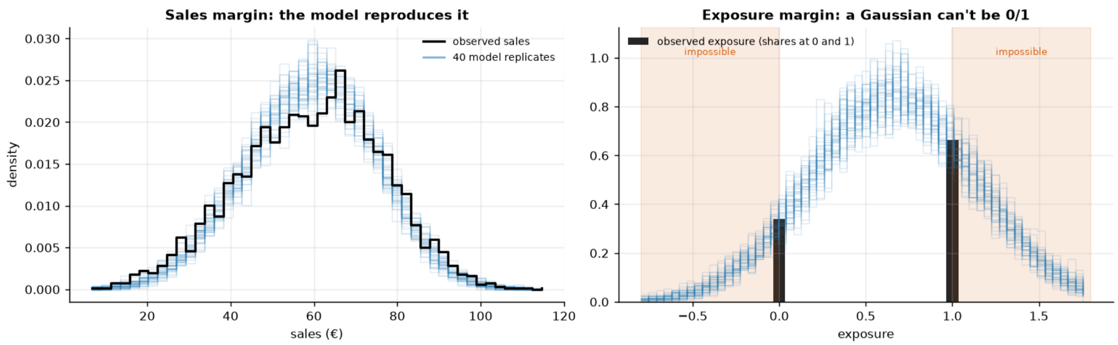
### 4b · Posterior predictive check — what the joint Gaussian gets right, and what it can't

Before leaning on the estimate we ask the fitted model to **reproduce the data it just saw**. For each of ~40 posterior draws we simulate a full replicate dataset from the fitted joint model,

$$(Y_i^{rep},\, X_i^{rep}) \;\sim\; \mathcal{N}_2\left( \begin{pmatrix} \hat{\beta}_0 + \hat{\beta}\, X_i \\ \hat{b}_0 + \hat{\pi}\, Z_i \end{pmatrix},\; \hat{\Sigma} \right),$$

(one $(\hat{\beta}_0, \hat{\beta}, \hat{b}_0, \hat{\pi}, \hat{\Sigma})$ per draw) and overlay the replicated margins on the observed ones. The **sales margin should pass** — that is what licenses reading the causal $\beta$ off the outcome equation. The **exposure margin cannot pass**: observed exposure is 0/1, and a Gaussian equation will cheerfully produce "exposures" of −0.3 or 1.4. We run it anyway, because that visible failure *is* the Step-3 modeling note, shown rather than footnoted.

```
Sales margin: replicate sd €15.7 vs observed €17.1, means €59.3 vs €59.3
  - close, with a mild under-dispersion (the joint approximation shows up
  even here, faintly): good enough to read beta off the outcome equation.
Exposure margin: 32% of replicated 'exposures' land outside [0,1]
  - impossible values, the linear-probability first stage made visible.
  Read that equation's point estimate and rho, never its width.
```



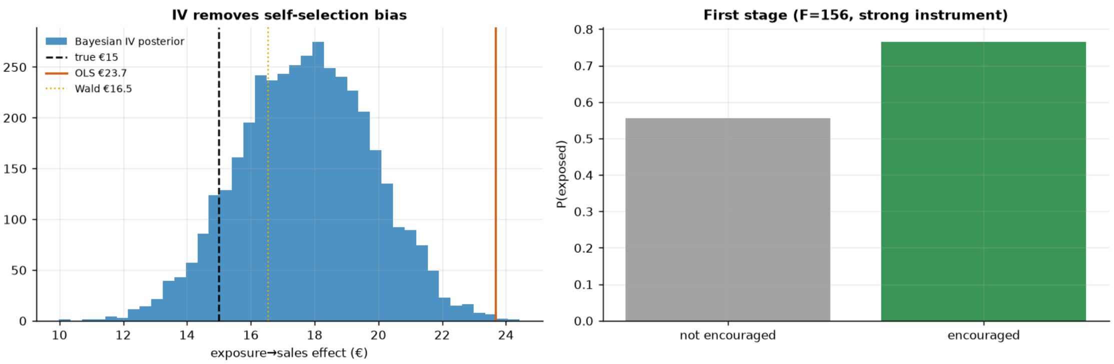

**Read-out.** *Left*: the replicate sales curves trace the observed distribution well — centred right, roughly the right width, if a hair narrower in the shoulders (the printed sd comparison quantifies it — even the "passing" margin carries a faint fingerprint of the approximation). That is a good enough fit to entitle us to read the causal $\beta$ (and its interval) off the outcome equation. *Right*: the replicated "exposures" are smooth bell curves, and a substantial share of their mass sits in the shaded regions below 0 and above 1 — exposure values no customer can have (the printed line quantifies it). This is not a surprise to fix; it is the **linear-probability approximation made visible**, and it is exactly why §5b will find a small residual bias and mild under-coverage, and why the Step-3 modeling note told us to use the exposure equation's point estimate and $\rho$ but never its posterior width. One diagnostic now carries what were two apologetic footnotes.

## 5 · Validate — IV removes the self-selection bias

Naive OLS should sit well *above* the truth; the Wald ratio and the Bayesian IV should both **remove that bias and land close to €15**, with the IV posterior's 90% interval **covering the truth**. The gap between OLS and IV **is** the self-selection bias the instrument removes. (We give the Bayesian IV weakly-informative priors so its interval reflects genuine sampling uncertainty — roughly the frequentist Wald width — rather than the artificially tight band CausalPy's default 2SLS-centred priors would produce.)

**One honest note on LATE (and who the compliers are).** IV recovers the effect for *compliers* — the customers whose exposure the encouragement actually moves — not the whole population. Under monotonicity that group is exactly the **first-stage shift: +21 pp (56% → 77%)**, the estimated complier share. In *this* simulation every customer's true effect is €15 (the effect is homogeneous), so **LATE = ATE** and grading the IV against €15 is a fair test. On real data, where the effect varies across customers, IV would recover the complier-weighted LATE, which generally **≠ the ATE** — and that is the right quantity for a budget call, since it answers *"what does an extra exposure do for the customers we can actually move?"*

```
self-selection bias removed: OLS €23.7 -> IV €17.7 (true €15)
```

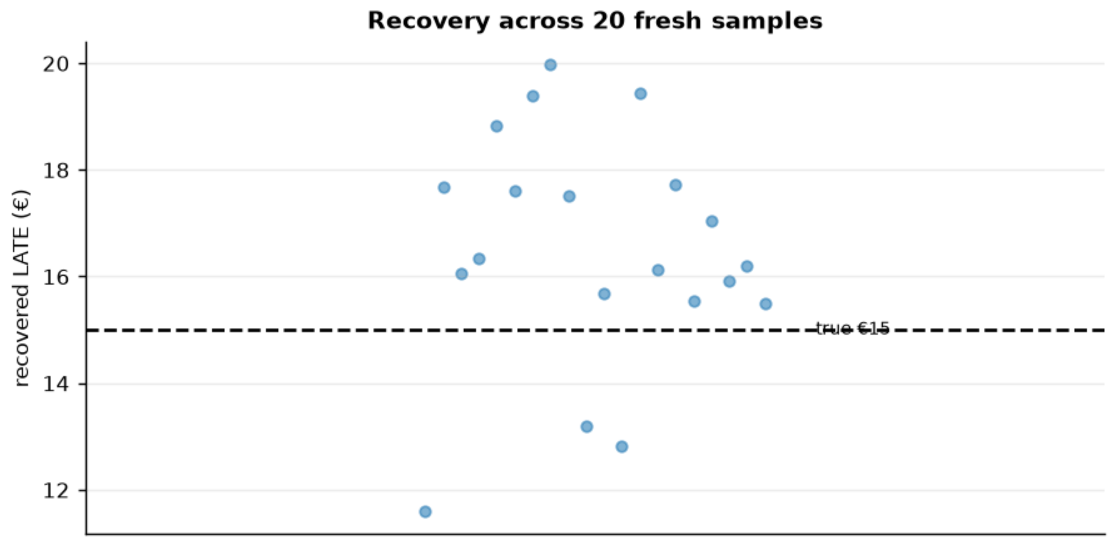



**How to read this.** *Left* — three vertical lines tell the whole story: **OLS (orange) sits well above the truth** (it credits the ad for pre-existing intent), while the **Bayesian IV posterior (blue) and the Wald ratio (gold) both drop back down toward the true €15**. The IV point carries a small upward finite-sample bias (a couple of euros above the truth on this seed; §5b measures the average across seeds), but its posterior — now honestly wide, thanks to weakly-informative priors — **covers €15** (just inside the lower edge); the artificially tight default-prior band would have excluded it. The horizontal distance from the orange line to the blue posterior *is* the self-selection bias — the euros of "effect" that were never causal. If you budgeted on OLS you'd over-spend on every exposure by exactly that gap. *Right* — the first stage: the encouragement clearly moves exposure (a big jump in P(exposed)), which is why the instrument is **strong** (F $\gg$ 10). A weak first stage here would make the whole IV estimate unstable, so this panel is a precondition, not a footnote.

### 5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test is whether the estimator is **centred on the truth over repeated samples** and whether its 90% interval **covers** the truth at the stated rate. We refit on many fresh samples (a small fast fit each; their sampler chatter is silenced below, and we read only the aggregate bias/coverage tallies) and check both.

```
IV across 20 seeds: mean €16.5 (true €15)  bias +1.5  sd €2.1
  *  90% interval covers truth in 16/20 seeds.
The committed seed (37) lands high (€18); across seeds the mean is €16.5
  - a small finite-sample bias of +1.5 € (~10%) toward OLS (the next cell asks
  whether it lives in the identification or in the model). The 90% interval
  covers €15 in 16/20 seeds (80% vs the 90% target - modest under-coverage,
  consistent with that bias); the budget verdict below never hinges on the
  last euro, which is why the ranking and the cap-style decision survive it.
```




**Is the residual bias the identification's fault, or the model's?** The Wald ratio *is* the IV identification with nothing else attached — two least-squares slopes and a division; no priors, no likelihood, no sampler. Re-running it on fresh draws at growing $n$ therefore isolates what finite samples do to the identification itself. Whatever bias it does **not** show must belong to the Bayesian machinery around it — and §4b already fingered the suspect: the Gaussian first stage.

```
closed-form Wald across 30 seeds (identification only, no model):
  n =  1,000:  mean € 14.4  (bias -0.6)   spread (sd) €3.71
  n =  3,000:  mean € 15.1  (bias +0.1)   spread (sd) €2.53
  n = 10,000:  mean € 15.0  (bias +0.0)   spread (sd) €1.43
The Wald column stays pinned near €15 at every n - its deviation is within
sampling noise while the spread shrinks like 1/sqrt(n). The identification is
clean; the posterior-mean bias above therefore belongs to the Bayesian model's
finite-sample behaviour (the Gaussian first stage §4b exposed, plus
posterior-mean-of-a-skewed-posterior at modest draw counts), not to the IV
logic. §5c prices the pure-IV leakage via the 1/F rule at a few cents.
```

### 5c · What if the instrument were weak? (the headline pitfall, demonstrated)

Everything so far used a **strong** instrument: the encouragement moves exposure by ~21 pp and the first-stage F sits in the hundreds. The caveat "weak instruments are dangerous" deserves better than assertion, so we now degrade the instrument on purpose and watch the estimator come apart.

**First, define the diagnostic we keep invoking.** For a single instrument, the first-stage F is the ordinary F-statistic of the regression $X = b_0 + \pi Z + u$ (exactly what `est.first_stage_F` computes):

$$F \;=\; \frac{R^2/1}{(1-R^2)/(n-2)},$$

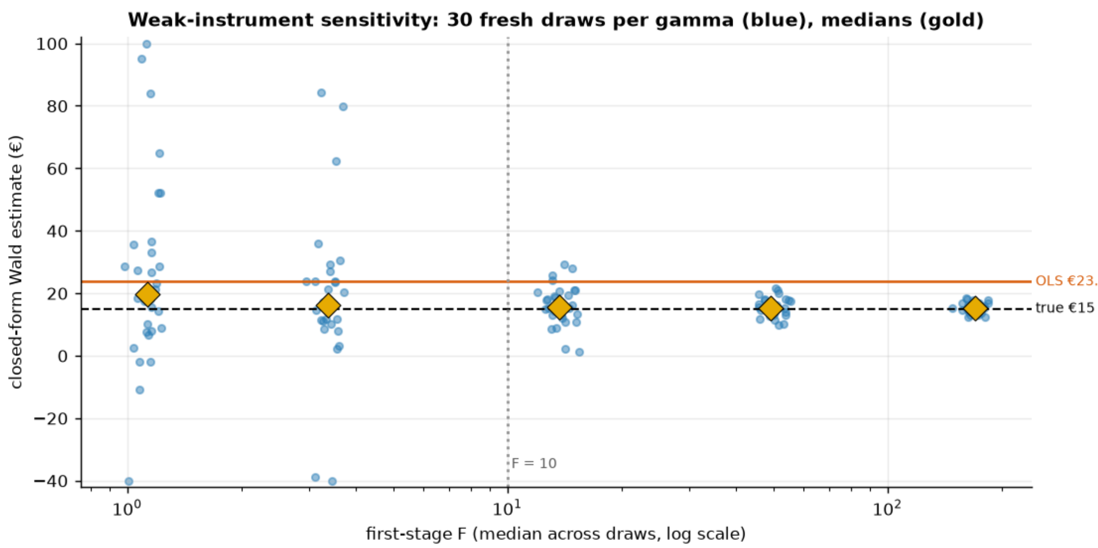
where $R^2$ is the share of exposure variance the instrument explains and $n-2$ counts the residual degrees of freedom: F measures the instrument's *signal* in the first stage relative to noise. **Why is the folklore threshold 10?** With one instrument, a standard approximation says the **2SLS** (two-stage least squares — the standard regression form of the same Wald ratio we computed above) / Wald estimator's residual bias, *relative to the OLS bias it is supposed to remove*, is roughly the reciprocal of F:

$$\frac{E[\hat{\beta}_{\text{2SLS}}] - \beta}{E[\hat{\beta}_{\text{OLS}}] - \beta} \approx \frac{1}{F},$$

so F = 10 means "no more than about a tenth of the OLS bias leaks back in" — a rule with teeth, not folklore. Cashed out on this notebook's numbers: F ≈ 156 prices the pure-IV leakage at €8.7/156 ≈ **€0.06** per exposure — negligible, which is why §5b's residual (more than an order of magnitude larger) had to come from the model approximation, not the instrument. (Caveat from `first_stage_F`'s own docstring: this is the classical **homoskedastic** F — it assumes the first-stage noise variance is constant across customers; on real, **heteroskedastic** data, where that variance changes across customers as real data usually does, prefer the Montiel Olea–Pflueger **effective F**, which corrects the F < 10 threshold for exactly that non-constant noise, and treat F > 10 as necessary, not sufficient.)

**The experiment.** `dgp.iv_ad_exposure` (src/cmp/dgp.py) hard-codes the encouragement coefficient at 1.1, so the cell below re-parameterizes the same six-line DGP with the first-stage strength $\gamma$ as a dial — $X \sim \text{Bern}(\sigma(0.3 + \gamma Z + 0.8U))$, $Y = 50 + 15X + 12U + \varepsilon$ — and sweeps $\gamma$ from this notebook's strength down to nearly-useless. For each $\gamma$ we draw many fresh datasets and compute the **closed-form Wald only** (two `lstsq` calls per draw — the failure we are demonstrating is the identification's, so no MCMC is needed and the cell stays seconds-fast).

```
  gamma  1.1: median F  169.6  median Wald € 15.2  90% range [12.7, 18.0]
  gamma  0.6: median F   49.3  median Wald € 15.3  90% range [10.7, 20.6]
  gamma  0.3: median F   13.7  median Wald € 15.6  90% range [5.0, 27.0]
  gamma 0.15: median F    3.4  median Wald € 16.3  90% range [-20.4, 71.9]
  gamma 0.08: median F    1.1  median Wald € 19.8  90% range [-6.9, 90.2]
At F ~ 1.1 the 90% range of the IV estimate spans €-7 to €90 - far wider than
the €8.7 OLS-vs-truth gap it was meant to fix, and the median drifts toward OLS
(€23.7): a weak instrument is worse than none - shown, not asserted.
(3 extreme point(s) clipped to the plot window.)
```



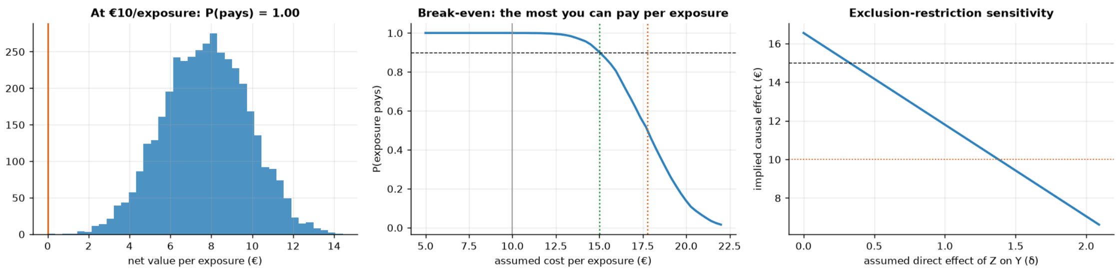

**How to read this.** Each blue column is the *same estimator on the same market* — only the instrument's strength changes. At the right edge (F in the hundreds, this notebook's regime) the Wald estimates hug the true €15. Moving left, the spread widens smoothly until, below the dotted F = 10 line, it explodes: single draws can land below zero or at multiples of the truth, and the gold *median* drifts up toward the orange OLS line. The mechanism is the 1/F formula read right-to-left — with a nearly-irrelevant instrument the Wald denominator is sampling noise around zero, so the ratio inherits OLS's bias *and* enormous variance. That is the precise sense in which **a weak instrument is worse than none**: OLS is wrong by a known-ish €9 with a tight interval, while a weak IV can be wrong by tens of euros in either direction *while wearing the costume of a causal estimate*. Lecture takeaway: check F before believing any IV number, and if F is small, walk away rather than report the ratio.

## 6 · Decide, in euros — and the exclusion-restriction stress test

Budget on the *causal* per-exposure value (IV), not the inflated OLS. But the exclusion restriction is untestable, so we ask: **what if the encouragement had a small direct effect on sales** (say the lottery email itself advertises)? We subtract a hypothetical direct effect $\delta$ from the reduced form and watch the implied causal estimate move — the honest bound on how much a leak would distort the number.

```
At €10/exposure: causal net €7.7  *  P(pays) 1.00
  -> BUY exposure (to the complier margin)
  P(pays)=1.00 is saturated, not assumed: the whole posterior
  [€14.3, €21.1] sits above €10.
Break-even sweep: exposure clears the 90%-confidence bar up to €15.0/exposure
  (the budget cap); at €17.8 it is a coin flip.
  So the €10 plan has ~€5.0 of headroom per exposure before the call gets tight.
Exclusion stress: the BUY call survives unless the instrument's direct effect
  exceeds delta~1.4 (out of a €3.5 reduced form) -
  trusting OLS (€24) would overstate the value and overspend.
```

```
(the corrected line is a point-estimate bound - it re-solves the Wald ratio
at posterior means; carry the IV interval's width alongside it when briefing
the cap.)
```




**Read-out.** Three numbers make the budget case, one per panel. *Left*: at the current €10/exposure the **causal** net value is ≈ €8/exposure with the entire posterior above zero — a saturated BUY, valid for the complier margin the instrument identifies. *Middle*: the sweep turns the posterior into a procurement cap — the 90%-confidence bar holds up to ≈ **€15/exposure**, the coin-flip point is ≈ €17.8, so the €10 plan has ~€5 of genuine headroom. *Right*: the stress test bounds the untestable worry — the BUY survives unless the lottery email itself drives more than $\delta \approx$ €1.4 of sales directly, which against a €3.5 total reduced form would be a large leak (a plain encouragement email with no product content makes that implausible; an email that doubles as an ad does not — design the instrument accordingly). The management contrast to land: budgeting on OLS's €24 would sanction paying up to €24 for something causally worth ≈ €15 — a €9 gap against the simulator's answer key. On real data the truth is unknown, so the wedge you'd actually avoid is OLS − IV ≈ €6/exposure (the IV prices the exposure at ≈ €17.7); either way, times every exposure bought, that is the price of mistaking correlation for incrementality — the cell below prices that same error at campaign scale (≈ €5.9M).

### The one-paragraph decision

> **What I'd tell the CMO.** Buy the exposure at the current €10: the causal value per exposure (the IV posterior above) sits comfortably clear of cost — P(pays) is saturated — and the break-even sweep caps procurement at ≈ €15 per exposure at 90% confidence, coin-flip near €18, so the €10 plan carries roughly €5 of headroom. **Scope:** that number is earned on the *complier margin* — the ≈ 21 pp of customers the encouragement actually tips into seeing the ad — and says nothing about always-exposed loyalists; that is the right scope for this call, because a budget buys *incremental* exposure. **Fragility, priced:** the BUY survives a direct-effect leak of the lottery into sales up to $\delta \approx$ €1.4 of the €3.5 reduced form (a large leak for a content-free lottery email), and survives the small finite-sample bias §5b measured; it would *not* survive a weak instrument — which is why F ≈ 156 was checked before anything else (§5c shows the crash we avoided). **The mistake we didn't make:** budgeting on the naive €23.7 would sanction paying several euros of non-causal value on every exposure bought — the cell below prices that error at campaign scale, and the JSON carries every number a deck needs.

```
Campaign-scale cost of the naive number: an OLS-calibrated cap sanctions
  ~ €5.9 of non-causal value per exposure; at 1,000,000 exposures/quarter that
```

```
  authorizes ~ €5.9M per quarter for value that does not exist
  (€2.5M-€9.3M against the IV 90% interval).
```


```
{
  "true_effect": 15.0,
  "ols": 23.67,
  "wald": 16.55,
  "iv_mean": 17.73,
  "iv_ci90": [
    14.34,
    21.15
  ],
  "first_stage_F": 156.2,
  "complier_share_pp": 21.1,
  "rho_mean": 0.218,
  "cost_per_exposure": 10.0,
  "budget_cap_c90": 15.02,
  "coinflip_c50": 17.77,
  "exclusion_break_delta": 1.43,
  "multiseed_bias": 1.5,
  "multiseed_coverage": "16/20",
  "naive_overpay_eur_per_quarter_at_1M": 5940601
}
```


## 7 · Caveats

- **LATE, not ATE.** IV speaks only for **compliers** — customers the encouragement moves (the Step-3 table shows who is silent). Always-exposed loyalists and never-exposed skeptics may respond differently; report the scope.
- **Weak instruments are dangerous.** A small first stage (low F) inflates both variance and bias — §5c demonstrated the crash: below F $\approx$ 10 the estimate's spread explodes and its median drifts back toward OLS. Check F first (on real data, the Montiel Olea–Pflueger *effective* F), and if it is small, walk away rather than report the ratio.
- **Exclusion is untestable.** We formalized it ($Y_i(z,x) = Y_i(x)$), stress-tested it in step 6, and priced the leak that would flip the call; defend it on design, not data.
- **Monotonicity (no defiers).** IV's LATE interpretation assumes no one is pushed *away* from exposure by encouragement — usually plausible, worth stating (it deleted the defier row in Step 3's derivation).
- **The joint Gaussian is an approximation.** §4b showed the exposure margin failing by construction (a 0/1 variable modeled linearly); §5b measured the resulting small bias and mild under-coverage. Read the exposure equation's point estimate and $\rho$, never its posterior width.
- **Real data next.** Notebook **11b** replays this playbook on the Criteo uplift dataset — randomized assignment as the instrument for actual ad exposure — where F, the complier share, and the budget cap must be earned from the data rather than planted in it.